Matching Techniques - Propensity Score Matching

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1 Introduction: Why Matching?

1.1 Policy Evaluation and Treatment Effect

In public policy research, we often aim to evaluate the effectiveness of a policy intervention. This type of study is called **policy evaluation**. The entities adopting the policy form the **treatment group**, while those not adopting the policy form the **control group**. Due to ethical and practical constraints, conducting **randomized experiments** in social sciences is difficult. As a result, we rely on **observational studies** and **quasi-experimental methods**.

1.2 Challenges: Selection Bias and Counterfactual Framework

Selection Bias:

- Selection bias arises when the initial conditions of the treatment and control groups are not completely comparable. Individuals may choose participation based on expected benefits, leading to biased treatment effect estimates.
- Example: In education policy, students who voluntarily attend additional training may be inherently more motivated, affecting their performance regardless of the training itself.

Counterfactual Framework:

- Introduced by Rubin (1974), the counterfactual framework provides a method to estimate causal effects.
- The key question in impact evaluation: *What would have happened to participants if they had not received the treatment?*

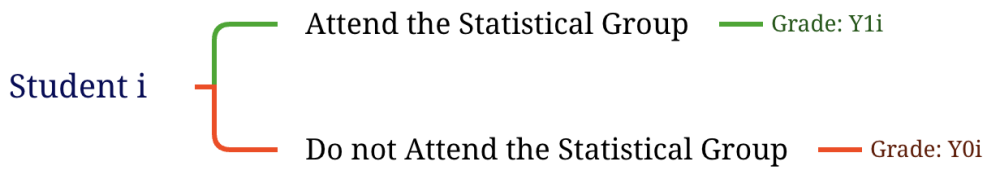


Figure 1: $Y_{1i} - Y_{0i}$ individual treatment effect

- **Potential Outcomes:** Y_{1i}, Y_{0i}
- **Counterfactual Outcomes:** After making a choice, the unobservable outcome is called the counterfactual outcome. It is the **potential outcome** that would have been observed in the absence of the intervention.

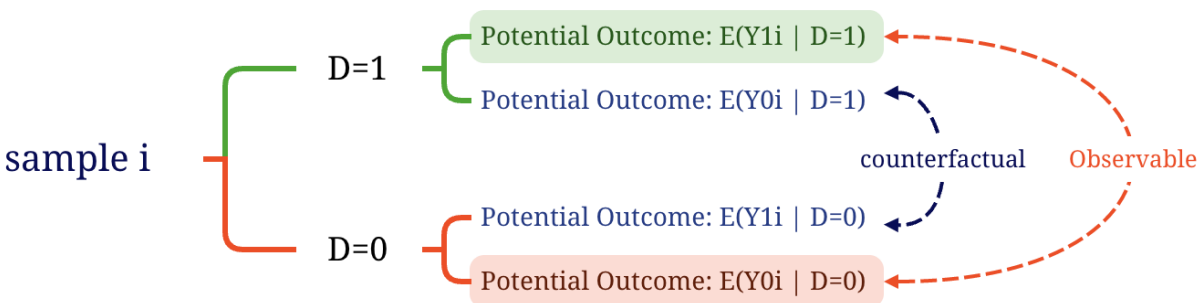


Figure 2

Example:

- If one chooses to attend ($D = 1$), then the grade without attending the statistical group (Y_{0i}) is the counterfactual outcome.
- If one chooses not to attend ($D = 0$), then the grade with attending the statistical group (Y_{1i}) is the counterfactual outcome.

$$E(Y_{1i}|D=1) - E(Y_{0i}|D=0)$$

$$= \underbrace{\boxed{E(Y_{1i}|D=1) - E(Y_{0i}|D=1)}}_{\text{ATT}} + \underbrace{\boxed{E(Y_{0i}|D=1) - E(Y_{0i}|D=0)}}_{\text{selection bias}}$$

Figure 3: selection bias

$$E(Y_{0i}|D=1)$$

The expected potential outcome for individuals who actually chose to receive treatment, **if they had not received treatment**.

$$E(Y_{0i}|D = 0)$$

The expected potential outcome for individuals who actually did not receive treatment, **if they had not received treatment**.

If:

$$E(Y_{0i}|D = 1) \neq E(Y_{0i}|D = 0)$$

Then an individual's choice to receive or not receive treatment is influenced by the **potential outcome**.

1.3 How to Address Selection Bias and Counterfactual Estimation?

- One solution is **matching estimators**, which construct a control group that is similar to the treatment group based on observable characteristics.
- By matching participants (treated) with non-participants (untreated), we can create a comparable counterfactual.

2 Why Use PSM for Matching?

In a general setting, x_i may include multiple variables, such as a K-dimensional vector. When matching x_i, it means performing matching in a high-dimensional space, which can lead to issues of data sparsity. In other words, it becomes difficult to find x_j that is sufficiently close to x_i.

To address this, a common approach is to apply a function $f(x_i)$ to compress the K-dimensional information into a single dimension, and then match based on $f(x_i)$.

One such method is the use of the **vector norm**, which defines a distance function in vector space. Considering the similarity or distance between x_i and x_j, the **Mahalanobis distance** is defined as:

$$d_M(x_i, x_j) = \sqrt{(x_i - x_j)^T S^{-1}(x_i - x_j)} \tag{1}$$

where S^{-1} is the inverse covariance matrix of the covariates.

2.1 Limitations of Mahalanobis Matching

- If the number of covariates is large, finding good matches becomes difficult.
- Even if two units are close in Mahalanobis distance, they may still differ significantly in absolute terms.
- To overcome this, **Rosenbaum and Rubin (1983)** proposed the **propensity score** as a new matching approach.

2.2 Conclusion

- PSM helps address selection bias in observational studies by creating comparable treatment and control groups.
- Instead of matching on multiple covariates, PSM uses a single score to balance groups effectively.
- While it improves causal inference, PSM still relies on strong assumptions and requires careful balance checks.

3 What is Propensity Score Matching

- Propensity Score Matching (PSM) is also known as Neyman-Rubin Causal Model.
- It is a technique used to create commonality among groups of observations/subjects across observed characteristics.
- It is used in observational research where random assignment is not feasible Reduces or eliminates sample selection bias in making comparisons across groups (e.g. treatment/control) on observable characteristics

3.1 Two Key Assumptions of PSM

- a. Conditional Independence Assumption (CIA) / Unconfoundedness

- The treatment assignment is independent of potential outcomes given the observed covariates:

$$Y_{1,i}, Y_{0,i} \perp D \mid x_1, x_2, \ldots, x_n$$

$$P(x_1, x_2, \ldots, x_n) = Pr(D = 1 \mid x_1, x_2, \ldots, x_n)$$

  The propensity score is the probability that an individual receives the treatment.

  Key condition: The explanatory variables must contain no information about the mean of unobserved factors.

- This implies that, after conditioning on covariates X, the treatment status does not provide additional information about the expected outcome.
- If unobserved confounders influence both treatment assignment and the outcome, this assumption is violated, leading to biased estimates.

- b. Common Support (Overlap) Assumption

- There must be sufficient overlap in the propensity score distribution between treated and untreated groups:

$$0 < P(T = 1|X) < 1 \tag{2}$$

- The loss of observations due to lack of common support can alter the representativeness of the sample, leading to sampling bias.

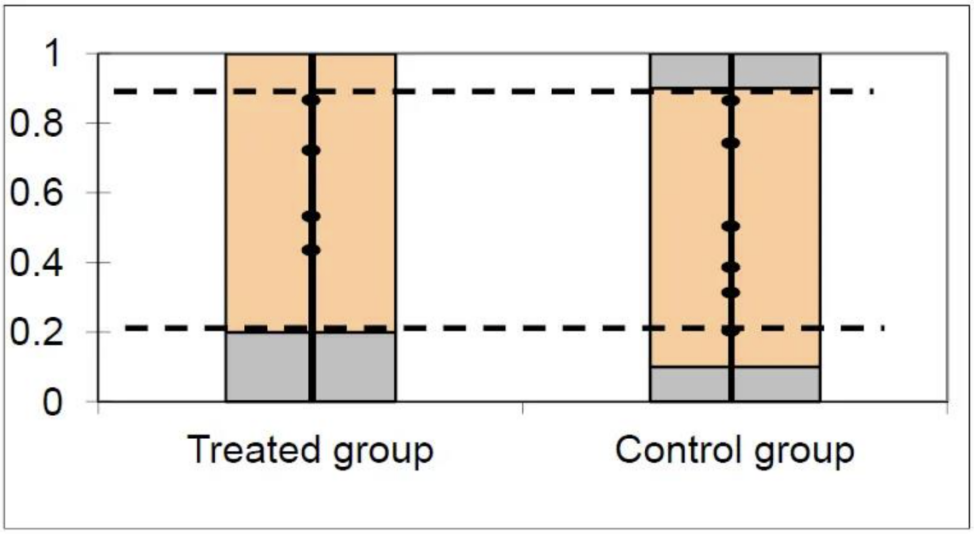


Figure 4: Matching with Common Support

- **Balancing Property**

- Propensity score matching ensures that the distribution of covariates is balanced between the treatment and control groups.
- Matching is not only done at the mean level but also balances the overall distribution of characteristics.
- If balance is not achieved, causal inferences from PSM may be biased.

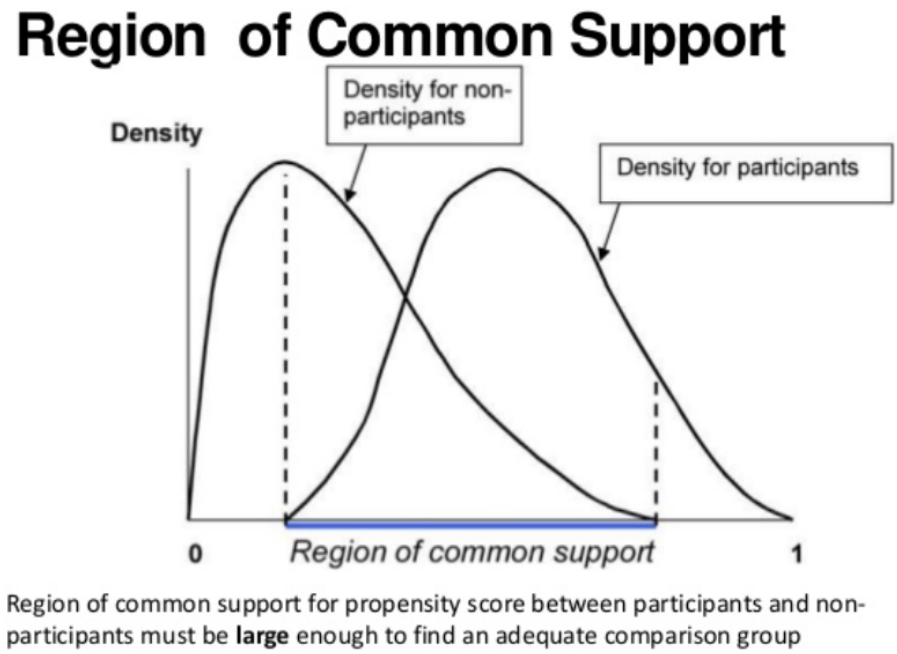
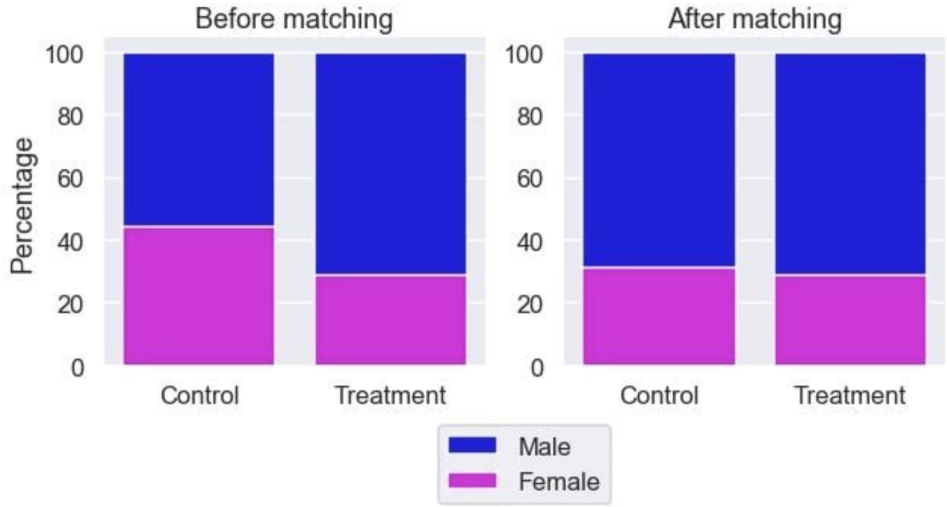
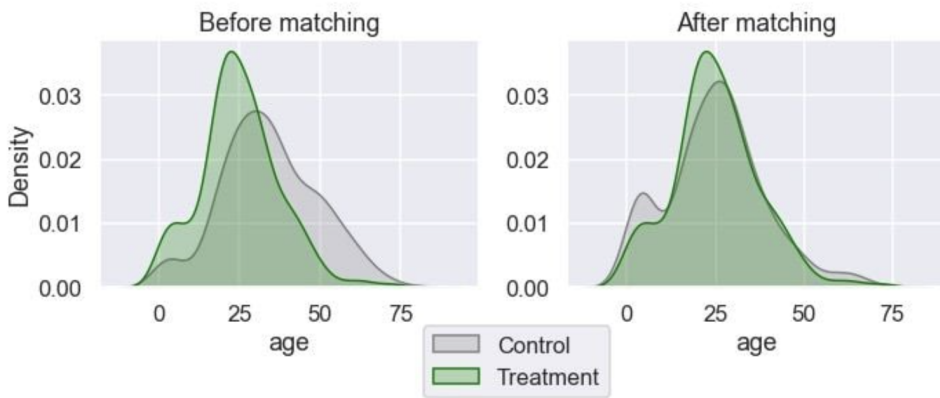


Figure 5: Common Support





4 How it works

1. **Select covariates:** Theoretically, selecting all variables that influence both an individual's probability of receiving the treatment and the outcome variable (Y) should be selected. But it is hard to implement.

   Rubin's perspective: 1. More influencing factors should be selected, especially those that affect the choice of treatment behavior. 2. The influencing factors (X_i) should affect an individual's decision to receive the treatment, rather than the treatment affecting the influencing factors (X_i). −¿ Only variables that occur before the treatment behavior can be selected as influencing factors.

   Others: Variables that are unrelated to the exposure but related to the outcome should always be included in PS model. In contrast, including the variables that are related to the exposure but not the outcome will decrease the precision of the estimated exposure effect without decreasing the bias. https://pubmed.ncbi.nlm.nih.gov/16624967/

2. **Estimate the propensity score**, usually using **logit regression**.

3. **Defining the Region of Common Support and Balancing Tests** - The region of common support ensures that treatment and control groups have overlapping distributions of propensity scores. - Balancing tests verify whether, within each quantile of the propensity score distribution, the **average propensity score and mean of covariates are the same** for treatment and control groups.

4. **Perform propensity score matching**, using one of the following techniques.

5. **Calculate the average treatment effect** (ATT, ATE, ATU) based on the matched sample.

   - **Exact Matching** - Each treated unit is matched only with control units that have exactly the same values for the selected covariates.

- **Nearest Neighbor Matching** - Matches each treated unit i with the control unit j that has the closest propensity score:

$$\min_j ||p(X_i) - p(X_j)||$$

  - Can be performed **with replacement** (a control unit can be matched multiple times) or **without replacement**. - **Trade-off**: Reduces bias but may increase variance if some control units are used multiple times.

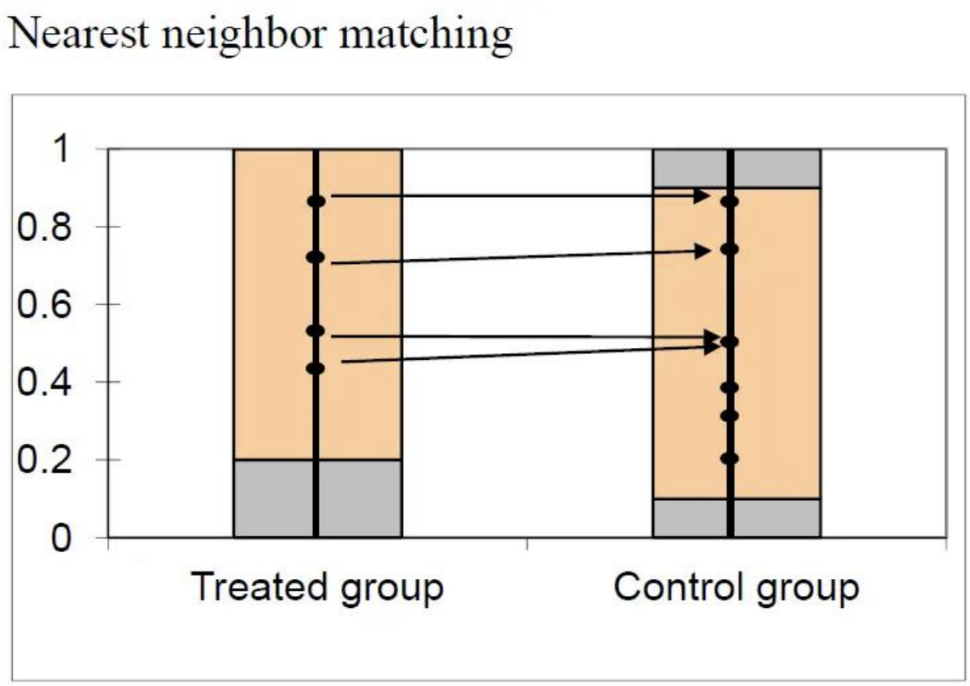


Figure 6

- **Kernel Matching** - Each treated unit i is matched with **multiple control observations**, with weights inversely proportional to the distance between treated and control units. - This smooth weighting reduces variance and increases efficiency. - Kernel weighting function ensures that closer control units receive larger weights.

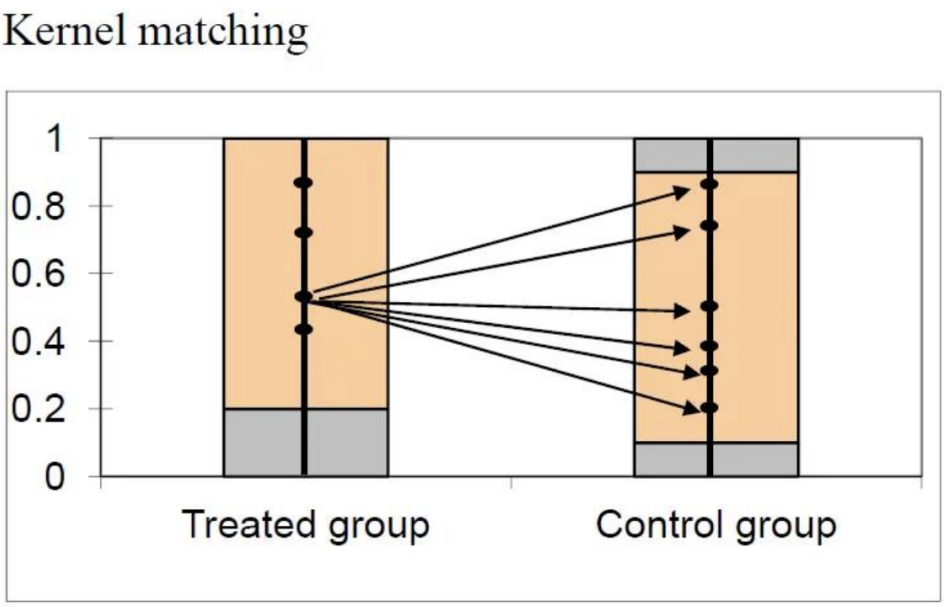


Figure 7

- **Caliper Matching (Radius Matching)** - Instead of matching to the single nearest neighbor, units are matched to all controls within a certain **radius** r:

$$A_i^k = \min_j \{p(X_i) | |p(X_i) - p(X_j)|| < r\}$$

- The choice of r is arbitrary but should be small to ensure comparability. - The treatment effect is estimated as:

$$E(Y_i^1 - Y_i^0 | p(X_i), T_i = 1) = \frac{1}{n^1} \sum_{i=1}^{\{T_i=1\}} \left[Y_i^1 - \frac{1}{k} \sum_{i=1}^{j \in A_i^k} w_{ij} Y_j^0 \right]$$

- Where w_{ij} is the relative weight of the matched control units.

5 ATE, ATT, ATU

Average Treatment Effect (ATE): The expected difference in outcomes between treated and untreated individuals in a population

$$E(Y_{1i} - Y_{0i})$$

Sample	
ATT	All treated individuals: $D = 1$ Average treatment effect of the treated group (ATT): $E(Y_{1i} - Y_{0i} \mid D = 1)$
ATU	All untreated individuals: $D = 0$ Average treatment effect of the untreated group (ATU): $E(Y_{1i} - Y_{0i} \mid D = 0)$

Individual	Treatment Status (D)	Potential Outcome Y1i	Potential Outcome Y0i	Individual Treatment Effect
Student 1	1	$Y_{1,1}$	$Y_{0,1}$	$Y_{1,1} - Y_{0,1}$
Student 2	1	$Y_{1,2}$	$Y_{0,2}$	$Y_{1,2} - Y_{0,2}$
Student 3	1	$Y_{1,3}$	$Y_{0,3}$	$Y_{1,3} - Y_{0,3}$
Student 4	1	$Y_{1,4}$	$Y_{0,4}$	$Y_{1,4} - Y_{0,4}$
...	...	...	...	...
Student 20	1	$Y_{1,20}$	$Y_{0,20}$	$Y_{1,20} - Y_{0,20}$
Student 21	0	$Y_{1,21}$	$Y_{0,21}$	$Y_{1,21} - Y_{0,21}$
Student 22	0	$Y_{1,22}$	$Y_{0,22}$	$Y_{1,22} - Y_{0,22}$
...	...	...	...	...
Student i	0	$Y_{1,i}$	$Y_{0,i}$	$Y_{1,i} - Y_{0,i}$

ATE

ATT

ATU

Figure 8

6 When to use to not to use PSM

- When randomized controlled trials (RCTs) are not feasible
- When you have observational data with selection bias
- When you need to reduce dimensionality in matching
- When the parallel trends assumption in DiD needs strengthening
- When there are many observed covariates affecting treatment assignment
- If there are unobserved confounders → PSM only balances observed variables. Consider instrumental variables (IV) or DiD.
- If treatment effects are highly heterogeneous → Standard PSM does not account for variation across subgroups.
- If the sample size is too small → Poor matching quality can lead to bias.

7 Limitations of PSM

(a) **A large sample size is usually required** to obtain high-quality matching results.

(b) **The common support assumption must hold**, i.e., the overlap in propensity scores between the treatment and control groups must be sufficiently large. Otherwise, observations outside this range will be discarded, leading to reduced sample representativeness.

(c) **PSM only controls for observed confounders**. If there are **unobserved confounders**, it can still introduce **"hidden bias."**